

DANIEL VARTANIAN

I'm a researcher at the University of São Paulo working on **complex systems** modeling and data science. My research focuses on **agent-based models**, geospatial analysis, and computational approaches to social and public health challenges.

I develop open-source research software, primarily in **R**, **Python**, and **NetLogo**. My tools are used by researchers studying chronobiology, food security, and climate impacts on health systems.

Outside research, I'm involved in building scientific communities and contributing to open-source projects through **rOpenSci** and the **NetLogo** ecosystem.



Access this CV online at:
danielvartan.github.io/cv-research

RESEARCH EXPERIENCE

Present
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2025

Research Fellow

Center for Metropolitan Studies (FFLCH-USP)

📍 São Paulo, SP, Brazil

- Lead geospatial modeling and large-scale data mining on the FAPESP-funded **AcessoSAN** project, investigating healthy food access in favelas and low-income urban communities across Brazil.
- Built reproducible **R** and **Quarto** pipelines to process and analyze millions of records from **SISVAN**, **Brazilian Revenue Service**, **DATASUS**, **IBGE Census**, and climate datasets like **WorldClim 2.1** and **Mapbiomas**.
- Collaborate with interdisciplinary teams of nutritionists, sociologists, health professionals, and engineers across multiple **USP** departments.

2025
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2024

Research Fellow

Sustentarea (FSP-USP)

📍 São Paulo, SP, Brazil

- Led agent-based modeling on the **CNPq**-funded **Global Syndemic** project investigating climate change effects on child nutrition and health in Brazil.
- Developed **NetLogo** models simulating food production systems and nutritional outcomes in children under 5, using climate projections from **Global Climate Models** and **IBGE** data.
- Created the **logolink** R package to provide analytical infrastructure for **NetLogo** simulations.
- Produced research papers (submitted), reproducible analysis pipelines, and technical reports for cross-institutional teams of health researchers and climate scientists from **USP** and the **RESICLIMA Network**.

2025
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2022

Graduate Research Fellow

University of São Paulo (USP)

📍 São Paulo, SP, Brazil

- Developed nested linear models investigating the relationship between human chronotype and latitude using Brazil's largest chronotype dataset (n=115,166).
- Created the **mctq** R package (published on **CRAN**, peer-reviewed by **rOpenSci**) for processing chronotype questionnaires.
- Built reproducible analysis pipelines in **R** and **Quarto** for statistical modeling and data processing.
- Completed **master's thesis** on chronotype and latitude relationships.

CONTACT

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in [danielvartan](#)

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REFERENCES

Prof. Dr. Aline Carvalho

Department of Nutrition
School of Public Health
University of São Paulo

Prof. Dr. Leandro Garcia

Senior Lecturer in Complexity
Science in Public Health
Centre for Public Health
Queen's University Belfast

Prof. Dr. Camilo R. Neto

Graduate Program in Complex
Systems Modeling
School of Arts, Sciences and
Humanities
University of São Paulo

*Developed with **R**
pagedown and datadrivencv
packages.*

*Source code available at:
github.com/danielvartan/cv-research*

Last updated on February 17, 2026.

2022
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2019

Assistant Research Fellow

University of São Paulo (USP)

📍 São Paulo, SP, Brazil

- Developed **actigraphy** monitoring protocols and analytical models for circadian rhythm assessment using non-parametric methods, cosinor analysis, periodograms, and actogram visualizations.
- Created the **actverse** R package providing tools for **actigraphy** data analysis.
- Co-authored the first **Brazilian Actigraphy Consensus** and delivered training workshops to 60+ researchers, including a launch workshop at the 18th Brazilian Sleep Congress.
- Contributed to USP's Sleep Genetics project investigating associations between **chronotype** and **clock genes** using **actigraphy** and genetic data from saliva samples across multi-centric cohorts in São Paulo and Baependi, Minas Gerais.

2019
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2016

Undergraduate Research Fellow

University of São Paulo (USP)

📍 São Paulo, SP, Brazil

- Provided statistical modeling and data analysis support for **circadian rhythm** and sleep research projects.
- Served as head of marketing and communications for USP's Interdisciplinary Sleep Research Group (**GIPSO**), securing media coverage in major Brazilian outlets like *Folha de São Paulo* and *TV Globo's Fantástico*.
- Co-developed sleep science courses through **EACH-USP's** summer program and the **Coursera** platform.



EDUCATION

2025
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2022

MSc, Complex Systems Modeling

University of São Paulo (USP)

📍 São Paulo, SP, Brazil

- Focus on agent-based and computational modeling of complex social and environmental systems.
- Founded USP's **Agent-Based Modeling Study Group**.
- Thesis: danielvartan.github.io/mastersthesis

2022
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2016

BSc, Marketing

University of São Paulo (USP)

📍 São Paulo, SP, Brazil

- Final grade: 7.7/10 (23rd/66)
- Strong quantitative foundation in calculus, statistics, and finance.
- Co-founded **EACH-USP's Wellness and Health Commission**.



TEACHING EXPERIENCE

2026
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2025

An Introduction to the R Programming Language

University of São Paulo (USP)

📍 São Paulo, SP, Brazil

- Developed and delivered introductory R programming course for researchers at the School of Public Health (**FSP**) (2025) and School of Philosophy, Literature and Human Sciences (**FFLCH**) (2026) at the University of São Paulo (**USP**).
- Trained approximately 30 researchers across both institutions.

PROGRAMMING LANGUAGES

🔗 [danielvartan](#)

Expert

📊 R

📖 NetLogo

Fluent

🐍 Python

📊 SQL

Intermediate

📄 Bash

📄 C C++ C#

📄 GAML (**GAMA**)

📄 JavaScript

📄 Lua

Basic Proficiency

📄 GDScript (**Godot**)

📄 Julia

📄 Rust

Markup/Other Languages

📄 CSS

🔗 Git

📄 HTML

📄 Markdown

📄 SASS

📄 TeX LaTeX

- 2024 • **Technical High School Program in Administration**
 São Paulo State Government  São Paulo, SP, Brazil
- Taught mathematics to approximately 100 students across two public high schools in São Paulo State's technical education program.
- 2023 • **ACH0042 - Problem Based Learning II**
 University of São Paulo (USP)  São Paulo, SP, Brazil
- Taught science and computational methods to 60 information systems students at EACH-USP.
- 2021 • **Introductory Course on Actigraphy**
 18th Brazilian Sleep Congress  São Paulo, SP, Brazil
- Co-delivered actigraphy training workshop accompanying the launch of the Brazilian Actigraphy Consensus.
 - Trained approximately 40 sleep researchers and health professionals.



SELECTED PUBLICATIONS

- 2025 • **Is Latitude Associated with Chronotype?**
 University of São Paulo (USP)
- Vartanian, D. (2025). *Is latitude associated with chronotype?* [Master's thesis, University of São Paulo]. <https://doi.org/10.11606/D.100.2025.tde-02042025-063648>
- 2023 • **Actigraphic Characterization of Sleep and Circadian Phenotypes of PER3 Gene VNTR Genotypes**
 Chronobiology International
- Viana-Mendes, J., Benedito-Silva, A. A., Andrade, M. A. M., Vartanian, D., Gonçalves, B. da S. B., Cipolla-Neto, J., & Pedrazzoli, M. (2023). Actigraphic characterization of sleep and circadian phenotypes of PER3 gene VNTR genotypes. *Chronobiology International*, 40(9), 1244–1250. <https://doi.org/10.1080/07420528.2023.2256858>
- 2021 • **Brazilian Actigraphy Consensus**
 Segmento Farma Editores
- Pedrazzoli, M., & Gonçalves, B. S. B. (Eds.) (with Benedito-Silva, A. A., Toscanini, A. C., Souza, A. P. P. B., Conway, B. A., Vartanian, D., Santos, E. H. R., Pires, M. L. N., & Leocadio-Miguel, M. A.). (2021). *Consenso brasileiro de actigrafia* [Brazilian actigraphy consensus]. Segmento Farma Editores. ISBN 978-65-992059-6-5.
- 2020 • **Early Chronotype with Advanced Activity Rhythms and Dim Light Melatonin Onset in a Rural Population**
 Journal of Pineal Research
- Ruiz, F. S., Beijamini, F., Beale, A. D., Gonçalves, B. D. S. B., Vartanian, D., Taporoski, T. P., Middleton, B., Krieger, J. E., Vallada, H., Arendt, J., Pereira, A. C., Knutson, K. L., Pedrazzoli, M., & Von Schantz, M. (2020). Early chronotype with advanced activity rhythms and dim light melatonin onset in a rural population. *Journal of Pineal Research*, 69(3). <https://doi.org/10.1111/jpi.12675>

AFFILIATIONS

Center for Metropolitan Studies
 Researcher (2025-Present)

RESICLIMA Network
 Researcher (2024-Present)

Sustentarea
 Researcher (2024-Present)

rOpenSci
 Software Developer (2021-Present)

Interdisciplinary Sleep Research Group (GIPSO)
 Researcher (2016-2022)

LANGUAGES

 danielvartan

English

 Fluent

 Fluent

 Fluent

 Fluent

Portuguese

 Native

 Native

 Native

 Native

Spanish

 Proficient

 Proficient

 Intermediate

 Intermediate

French

 Beginner

 Beginner

 Beginner

 Beginner



SELECTED SOFTWARE PROJECTS

- 2025

logolink

Creator and Maintainer

R package providing a **tidy** interface between R and **NetLogo**.

 - Published on **CRAN** (1,200+ downloads).
 - Currently under **peer review** by **rOpenSci**.
- 2025

LogoScope

Creator and Maintainer

VS Code extension providing syntax highlighting for **NetLogo**.

 - Published on **Visual Studio Code Marketplace** (227 installs) and **Open VSX** (842 downloads).
- 2025

LogoClim

Creator and Maintainer

NetLogo model developed to simulate and visualize climate conditions using data from **WorldClim 2.1**.

 - Published on **CoMSES Net**.
 - Currently under **peer review** in **JOSS** (2025).
- 2023

actverse

Creator and Maintainer

R package providing a **tidy** toolbox for **actigraphy** data analysis.
- 2021

mctq

Creator and Maintainer

R package providing a **tidy** toolbox for processing the Munich ChronoType Questionnaire (**MCTQ**).

 - Published on **CRAN** (13,000+ downloads).
 - **Peer reviewed** by **rOpenSci**.
 - Cited in chronobiology research across multiple countries.



SELECTED OUTREACH PROJECTS

- Present
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2025

Interdisciplinary Brazilian Symposium on Agent-Based Models (SIMBRA)

University of São Paulo (**USP**) 📍 São Paulo, SP, Brazil

 - Co-founded SIMBRA as an extension of USP's **Agent-Based Modeling Study Group** to provide a national platform for researchers and students across disciplines to discuss and disseminate agent-based modeling approaches.
- 2018
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2017

Durma na EACH (Sleep at EACH)

University of São Paulo (**USP**) 📍 São Paulo, SP, Brazil

 - Created sleep and rest spaces for students at **EACH-USP** with plans for expansion to São Paulo public schools.